

Table 1

Representative enzyme scaffolds for the elongation or assembly of natural and modified nucleic acids.								
Application	Enzyme family / scaffold	EC number	Database size ^[a]	Thermophilic origin ^[b]	Catalytic machinery	Representative enzyme	Representative structures ^[c]	Ref. ^[d]
Template-independent oligonucleotide synthesis	DNA polymerase (Family X)	2.7.7.31	8 920	23	2x Mg/Mn ²⁺ coordinated by 3 conserved Asp	Murine TdT (UniProt: P09838)	4I2F, 4QZ9, 4QZF	[38]
		2.7.7.7				Tardigrade RvPoIX (UniProt: A0A1D1UV65)	-	[39]
	Poly(U) RNA Polymerase	2.7.7.52	14 046	0		<i>S. pombe</i> Cid1 (UniProt: O13833)	4FHV, 4FH5, 4EP7, 4NKU	[40]
	Poly(A) RNA Polymerase	2.7.7.19	11 499	1		Bovine poly(A) polymerase (UniProt: P25500)	1F5A, 1Q79	[41]
	Primase-Polymerase (Archaeo-Eukaryotic)	2.7.7.7	7976	70	2x Mg/Mn ²⁺ coordinated by 2 conserved Asp	Mimivirus PrimPol (UniProt: Q5UQR0)	-	[42]
	DNA polymerase (Family A)		68 570	1760	2x Mg/Mn ²⁺ coordinated by 2 conserved Asp/Glu	Human Polθ (UniProt: O75417)	7ZUS, 7ZX1, 5AGA	[42]
Template-dependent oligonucleotide synthesis	DNA polymerase (Family A)	2.7.7.7	70 615	2188	2x Mg/Mn ²⁺ coordinated by 2 conserved Asp	Bst DNA polymerase (UniProt: E1C9K5)	4YFU, 6DSW, 6MU4, 7K5S	[43]
	DNA polymerase (Family B)					Thermostable SFM4-3 (UniProt: P52028)	8XJR, 8XK9, 9FM3, 9FMF	[44]
	RNA polymerase (single subunit)	2.7.7.6	7489	3	2x Mg/Mn ²⁺ coordinated by 2 conserved Asp + nucleophilic Lys	Thermostable Tgo (UniProt: P56689)	9MYE, 9GP8, 2XHB	[45]
						T7 RNA polymerase (UniProt: P00573)	1MSW, 1CEZ, 1QLN, 1H38	[46]
Oligonucleotide ligation (template-independent or dependent)	DNA ligase (ATP-dependent)	6.5.1.1	1407	3	2x Mg/Mn ²⁺ coordinated by 2 conserved Asp + nucleophilic Lys	T4 DNA Ligase (UniProt: P00970)	6DT1, 6DRT, 5WFF	[47]
						T3 DNA ligase (UniProt: P07717)	-	[48]
						R2D ligase (UniProt: M1EZJ7)	-	[49]
						<i>E. coli</i> DNA Ligase (LigA) (UniProt: P15042)	2OWO, 4GLX, 5TT5	[50]
	DNA ligase (NAD-dependent)	6.5.1.2	33 855	1313	2x Mg/Mn ²⁺ coordinated by conserved Asp + nucleophilic Lys	Taq DNA Ligase (UniProt: Q9ZFY7)	-	[51]
	RNA ligase 1 (ATP-dependent)	6.5.1.3	771	5		T4 RNA ligase 1 (Rnl1) (UniProt: P00971)	5TT6, 2C5U, 9DO4	[52]
	RNA ligase 2 (ATP-dependent)		698	7		T4 RNA ligase 2 (Rnl2) (UniProt: P32277)	1S68, 2HVQ, 2HVR, 2HVS	[53]
	RNA ligase 3 (ATP-dependent)		409	91		Thermophilic RNA ligase (PpaRnl) (UniProt: A0A075LQ94)	8UCF, 8UCG	[54]
	RNA splicing ligase (GTP-dependent)	6.5.1.8	18 643	826	2x Mg/Mn ²⁺ coordinated by Asp/Cys/His + nucleophilic His	RNA-splicing ligase RtcB (UniProt: P46850)	-	[55]

Within our webapp, users can further refine these datasets by filtering according to protein length, sequence identity relative to the experimentally characterized seed sequences, and extremophilic annotations (such as thermophilic origin) retrieved from the BacDive[64] database, thereby enabling targeted exploration of sequence space.

[a] Based on MMseqs2 easy-search with representative enzymes as seed collecting homologs with a minimal sequence identity of 25%.

[b] Based on BacDive annotation (growth temperature ≥ 45 °C).

[c] When applicable, selected structures display high resolution or substrate/mechanistic relevance.

[d] Related to the representative enzymes highlighted in this review.
